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Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam – 603 110

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Computer Science and Engineering

Continuous Assessment Test – I

Regulations – R2023

Degree & Branch	M.Tech & Computer Science and Engineering			Semester	V
Subject Code & Name	ICS1503 & Web Application Development				
Academic Year	2025-2026 ODD	Batch	2023-2028	Date	20-08-2025
Time: 08:10 - 09:40 a.m (90 Minutes)	Answer All Questions			Maximum: 50 Marks	

(K1: Remembering, K2: Understanding, K3: Applying, K4: Analyzing, K5: Evaluating)

CO1:	Make use of HTML5 and CSS3 to design modern web site (K3)
CO2:	Utilize Javascript and DOM to implement dynamic web page (K3)
CO3:	Develop web applications using Servlets (K3)

Part – A (4 × 2 = 8 Marks)

		KL	CO	PI
1.	List any <i>three text-related CSS properties</i> and provide an example usage for each.	K1	CO1	1.4.1
2.	Explain the differences between the <div> and elements in HTML.	K2	CO1	1.4.1
3.	Compare the <i>CSS units em and rem</i> , explain usage with suitable examples.	K2	CO2	1.4.1
4.	Why is the <i>XML configuration file</i> important in a <i>servlet application</i> ?	K1	CO3	1.4.1

Part – B (3×6 = 18 Marks)

		KL	CO	PI
5.	Explain <i>CSS positions</i> with an example.	K2	CO1	1.4.1
6.	Develop a <i>JavaScript program</i> that contains at <i>least three <p> (paragraph) elements</i> in an HTML page. When the user <i>clicks on the third paragraph</i> , the <i>first paragraph's background color and text</i>	K3	CO2	1.4.1 13.3.1

	color should change, and an image should be displayed on the page. When the user double-clicks on the third paragraph, the first paragraph should revert to its original background color and text color, and the image should disappear.			
7.	Explain <i>Interfaces and Classes</i> in <i>javax.servlet</i> package	K2	CO3	1.4.1

Part – C (2 × 12 = 24 Marks)

		KL	CO	PI												
8.	Design the following table using <i>HTML <table> elements and necessary attributes</i> to structure. Explain the different <i>types of CSS declarations and selectors</i>. Illustrate your explanation with the same design code that uses multiple types of selectors and shows how <i>specificity affects</i> which styles are applied. <table><tr><td>Quick</td><td>brown fox</td><td>jumps</td></tr><tr><td></td><td>lazy dog</td><td>and</td></tr><tr><td>over the</td><td>then it</td><td>fall</td></tr><tr><td></td><td>prey to a lion</td><td></td></tr></table>	Quick	brown fox	jumps		lazy dog	and	over the	then it	fall		prey to a lion		K3	CO1	1.4.1 2.2.3 13.3.1
Quick	brown fox	jumps														
	lazy dog	and														
over the	then it	fall														
	prey to a lion															
(Or)																
9.	Design a <i>CSS animation for an image element</i> that makes the image rotate continuously with a bounce effect. Use <i>CSS keyframes and appropriate animation properties</i> to achieve this. Explain difference between <i>animation fill mode and animation direction</i>.	K3	CO1	1.4.1 2.2.3 13.3.1												
10.	Develop a memory match game using <i>HTML, CSS, and JavaScript</i> with a <i>3x3 grid of boxes</i> containing <i>pairs of fruit names</i> and one special "<i>danger</i>" box. The danger box adds an extra challenge by immediately ending the game and resetting the score if clicked, encouraging players to be cautious while selecting boxes. <i>All boxes start hidden, and clicking reveals their content with matched pairs staying visible and assigned a shared random background color, awarding 10 points per match. Non-matching pairs are hidden again after a short delay, and the score resets to zero. The game ends with a success message when all pairs are matched, and the current score should be displayed prominently throughout the game.</i>	K3	CO2	1.4.1 2.2.3 3.4.2 13.3.1												
(Or)																
11.	Construct a <i>dynamic image gallery using JavaScript and the DOM</i> that displays five images, each with a description. The initial <i>HTML</i>	K3	CO2	1.4.1 2.2.3												

	body should contain only two empty <code><div></code> elements, one for the image and one for the description. Using JavaScript, dynamically create and insert all necessary elements including the image, description text, and Next and Previous buttons. The buttons should allow the user to navigate through the images one at a time, updating the displayed image and description dynamically on each click.			3.4.2 13.3.1
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Computer Science and Engineering

Continuous Assessment Test – II

Regulations – R2023

Degree & Branch	M.Tech & Computer Science and Engineering				Semester	V
Subject Code & Name	ICS1503 & Web Application Development					
Academic Year	2025-2026 ODD	Batch	2023-2028	Date	06-10-2025	FN
Time: 08:10 - 09:40 a.m (90 Minutes)	Answer All Questions				Maximum: 50 Marks	

(K1: Remembering, K2: Understanding, K3: Applying, K4: Analyzing, K5: Evaluating)

CO3:	Develop web applications using Servlets (K3)
CO4:	Develop micro services using Spring Boot (K3)
CO5:	Develop full stack web applications using Vue JS and Spring Boot (K6)

Part – A (4 × 2 = 8 Marks)

		KL	CO	PI
1.	Explain about Request Dispatcher ().	K2	CO3	1.4.1
2.	List differences between Servlet and Spring Boot	K1	CO4	1.4.1
3.	What is Vue?	K1	CO5	1.4.1
4.	Explain Reactive State?	K2	CO5	1.4.1

Part – B (3×6 = 18 Marks)

		KL	CO	PI
5.	Develop an application to implement an Image Slider using Vue.	K3	CO5	1.4.1 2.1.2
6.	Develop a simple To-Do List application using Vue where a user can add tasks, mark them as completed, and remove tasks from the list.	K3	CO5	1.4.1 2.1.2
7.	Explain the following spring boot annotations with examples: <i>Response Entity, Rest Controller and Request Mapping.</i>	K2	CO4	1.4.1

Part – C (2 × 12 = 24 Marks)

		KL	CO	PI
8.	(a) Construct a <i>backend system using Servlets to verify user login credentials</i> . The system should check the credentials entered by the user against a MySQL database. <i>If the credentials are valid, redirect the user to the home page; otherwise, redirect the user back to the login page.</i> (b) Explain in detail <i>the session management methods</i> used in a Servlet application.	K3	CO3	2.2.3 3.4.2 13.3.1
(Or)				
9.	(a) Develop a <i>backend system using Servlets to retrieve users whose age is greater than 45, update their job experience details (e.g., 10 years), save the changes in the database, and display the updated information.</i> (b) Explain in detail <i>the Cookie methods</i> used in a Servlet application.	K3	CO3	2.2.3 3.4.2 13.3.1
10.	Explain the process of <i>integrating database (MongoDB/ PostgreSQL) access into a Spring Boot application</i> along with examples.	K2	CO4	2.2.3 3.4.2 13.3.1
(Or)				
11.	Explain <i>REST API and the MVC architecture in a Spring Boot application</i> with examples and provide a <i>sample Controller demonstrating all HTTP verbs</i> .	K2	CO4	2.2.3 3.4.2 13.3.1